

Korsakoff syndrome (KS) is caused by a thiamin (vitamin B₁) deficiency that often results from chronic alcohol consumption. If KS is diagnosed during its earliest stages, its course may be reversed with thiamin supplements; otherwise, it progresses into a permanent type of dementia that includes severe cognitive dysfunction and memory loss. Alzheimer disease (AD) is associated with a different type of dementia that worsens over time, eventually causing memory loss, cognitive dysfunction, and severe brain atrophy.

In one study, KS and AD subjects were assessed using the Corsi block-tapping task, which measures nonverbal memory. The experiment consisted of 10 KS subjects, 10 AD subjects, and 10 healthy control subjects. The control group was matched by age, gender, and premorbid IQ (intelligence quotient preceding the occurrence of symptoms) to the KS and AD groups.

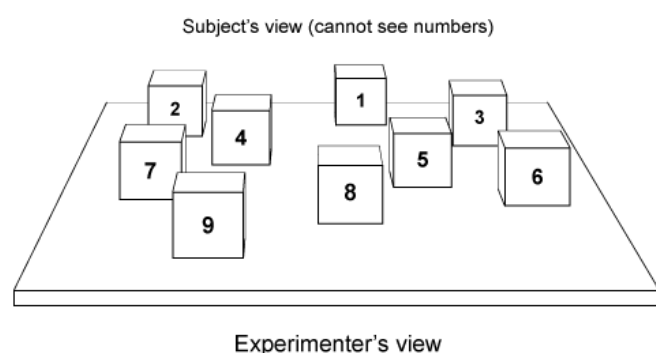


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Results indicated that compared to controls, subjects with AD and KS were significantly impaired on the delayed memory task, although all subjects showed a significantly greater number of errors with increases in the delay interval.

Which of the following best summarizes a difference between Korsakoff Syndrome (KS) and Alzheimer Disease (AD)?

- ☐ A. KS has a physiological cause, but AD does not. [2%]
- ☐ B. KS is caused by environmental insult, but AD is genetic. [10%]
- ☐ C. KS is associated with biochemical changes in the brain, but AD is not. [1%]
- ☒ D. KS can be reversible, but AD is irreversible. [86%]

Correct
85% answered correctly

Explanation:

Korsakoff syndrome	
Symptoms	• Memory loss • Confabulation (memory fabrication without the intention to lie)
Causes	• Extreme thiamine (vitamin B₁) deficiency resulting from: ◦ Chronic alcohol consumption (most common) ◦ Eating disorders &/or chronic malnutrition
Treatment/prognosis	• Vitamin B₁ supplementation can reverse most/all symptoms if caught early

The umbrella term **dementia** describes a decline in one or more aspects of **cognitive functioning** (eg, memory, attention, executive functioning,

perception, motor organization, language, social cognition). Declines result in poorer social, occupational, and/or daily functioning and can be associated with personality changes and difficulty regulating emotions. Although dementia can be caused by a range of diseases or brain injury, dementia due to Alzheimer disease (AD) is the most common type.

Dementia due to **Korsakoff syndrome** (KS) is caused by a **deficiency in thiamin** (vitamin B₁), typically resulting from chronic alcohol consumption (alcohol use disorder). People with alcohol use disorder generally have poor diets, and alcohol interferes with the absorption of thiamin in the digestive tract.

Thiamin, an essential nutrient obtained from one's diet, is necessary for proper neural functioning. Deficiency leads to cell death, particularly in the thalamus, the primary relay station for sensory information en route to the cerebral cortex, and the mammillary bodies, which have extensive connections with the hippocampus (memory formation).

Thiamin supplementation and behavior modification in the early stages of KS can potentially reverse most brain damage, but AD is irreversible, ultimately progressing to global cell death and severe brain atrophy.

(Choice A) KS and AD both have physiological components; AD is caused by abnormal beta-amyloid plaques and neurofibrillary tangles that interfere with proper neural signaling and result in cell death.

(Choice B) KS and AD both have environmental as well as genetic components. KS results from excessive alcohol consumption and poor nutrition (environmental), but alcohol use disorder appears to be at least partially heritable (genetic). Risk factors for AD include a mutation causing greater amyloid plaque accumulation (genetic) and many lifestyle factors (environmental).

(Choice C) KS and AD are both associated with biochemical changes in the brain that ultimately result in neuronal cell death.

Educational objective:

Dementia is an umbrella term that reflects a decline in one or more aspects of cognitive functioning. Dementia due to Korsakoff syndrome (KS) is caused by thiamin deficiency and is most commonly the result of chronic

alcohol consumption. Thiamin supplementation during the early stages of KS can reverse most symptoms of KS.

Time spent:0 sec

Subject:

Behavioral Sciences

QID:400896

System:

Learning, Memory, and Cognition

Korsakoff syndrome (KS) is caused by a thiamin (vitamin B₁) deficiency that often results from chronic alcohol consumption. If KS is diagnosed during its earliest stages, its course may be reversed with thiamin supplements; otherwise, it progresses into a permanent type of dementia that includes severe cognitive dysfunction and memory loss. Alzheimer disease (AD) is associated with a different type of dementia that worsens over time, eventually causing memory loss, cognitive dysfunction, and severe brain atrophy.

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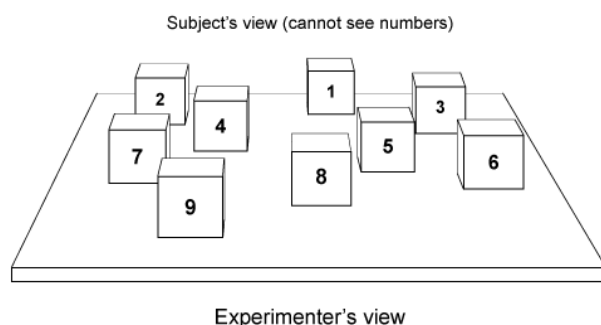


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Which of the following cognitive abilities is critical to the Corsi block-tapping task

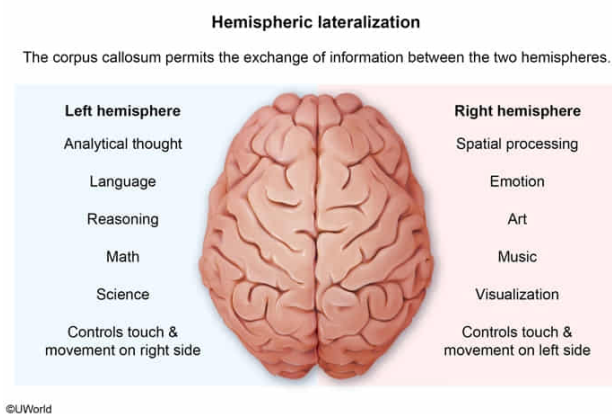
and would be MOST affected by a lesion to the right hemisphere?

- ☐ A. Emotional processing [5%]
- ☐ B. Comprehension of verbal instructions [10%]
- ✓ ☒ C. Visuospatial skills [76%]
- ☐ D. Analytical reasoning skills [7%]

Correct

76% answered correctly

Explanation:



The brain is divided into right and left hemispheres, each of which is **specialized** for certain processes; this specialization is known as **hemispheric lateralization**. Each hemisphere is responsible for **contralateral control** of the body: the left hemisphere controls touch and movement on the right side of the body, and vice versa.

The **right hemisphere** is crucial for processing **visuospatial** patterns, such as those required for the Corsi block-tapping task; deficits in visuospatial processing are often indicative of damage to the right hemisphere. In addition to visuospatial processing, the right hemisphere is important for **artistic** and **musical** abilities, visualization, and **emotion**.

(Choice A) Although the right hemisphere *is* specialized for processing emotion, that cognitive ability is not critical to the Corsi block-tapping task.

(Choices B and D) The **left hemisphere** is important for **language** ability, including speaking, writing, and comprehension. The left hemisphere is also superior at **analytical reasoning** skills and logic, which are used in mathematics and science. The Corsi block-tapping task relies on comprehension of verbal instructions, but not on analytical reasoning skills.

Educational objective:

Hemispheric lateralization is the specialization of each brain hemisphere for certain cognitive functions. The right hemisphere is specialized for visuospatial, emotional, and artistic/musical processing. The left hemisphere is specialized for linguistic and analytical processing.

Time spent:0 sec**Subject:**

Behavioral Sciences

QID:400897**System:**

Learning, Memory, and Cognition

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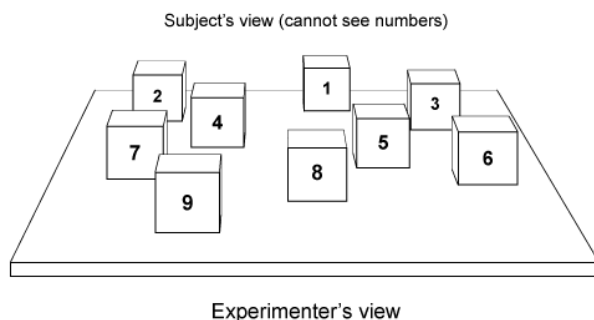


Figure 1 Corsi block-tapping task

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Performance of which of the following study tasks utilizes the mirror neuron

system?

I. Immediate recall task

II. Delayed recall task

III. Distractor task

☐ A. I only [14%]

☐ B. III only [14%]

☐ C. II and III only [7%]

✓ ☒ D. I, II, and III [63%]

Correct

62% answered correctly

Explanation:

Studies suggest that **mirror neurons** are specialized neurons in the brain that **fire both** while **observing** and while **performing** a behavior. Mirror neurons have been observed in various brain regions (eg, frontal and parietal lobes) of humans, other primates, and some birds, and are hypothesized to play a central role in observational learning.

Observational learning (also called modeling) describes when an individual learns how to perform a behavior by watching someone else perform that behavior. Watching and replicating movements, such as during the Corsi block-tapping task, is a form of observational learning, which is thought to utilize the mirror neuron system.

(Number I) As part of the Corsi block-tapping task, the immediate recall task requires subjects to touch the same blocks as the experimenter immediately afterward, which would rely on the mirror neuron system.

(Number II) The delayed recall task requires subjects to touch the same blocks the experimenter touched after a delay of 5–25 seconds; this is an example of observational learning and would also rely on the mirror neuron system.

(Number III) The distractor task requires subjects to watch and reproduce a finger-tapping sequence first demonstrated by the experimenter, which would also rely on the mirror neuron system.

Educational objective:

Located in various areas of the brain, mirror neurons are specialized neurons that fire both while observing and performing a behavior. Mirror neurons are

hypothesized to play a central role in observational learning.

Time spent:0 sec

Subject:
Behavioral Sciences

QID:400898

System:
Learning, Memory, and Cognition

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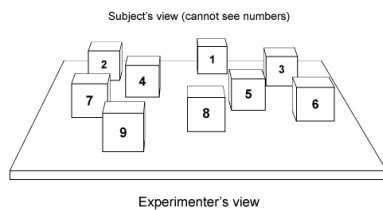


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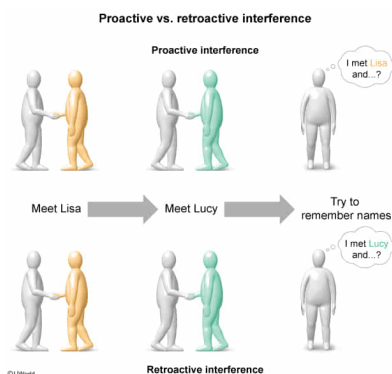
Results indicated that compared to controls, subjects with AD and KS were significantly impaired on the delayed memory task, although all subjects showed a significantly greater number of errors with increases in the delay interval.

The distractor and Corsi block-tapping tasks are designed to assess which of the following?

- ☐ A. Proactive interference and sensory memory recall [13%]
- ☐ B. Proactive interference and decay in episodic memory [8%]
- ☒ C. Retroactive interference and short-term memory recall [75%]
- ☐ D. Retroactive interference and decay in semantic memory [2%]

Correct
75% answered correctly

Explanation:



During the Corsi block-tapping task, the experimenter touches a sequence of blocks. The subject must then recall the exact sequence performed by the experimenter, after either no delay or a short (5–25 second) delay, during which the subject performs a distractor task. In other words, the subject must hold the information in **short-term memory**, which has a duration of 15–30 seconds. Therefore, the Corsi block-tapping task is a measure of *visuospatial short-term memory*.

Exposure to a distraction can interfere with the ability to accurately retrieve a memory or learn new information. **Retroactive interference** occurs when more **recently learned** information **interferes** with the learning of older information. In this study, the distractor task *retroactively interfered* with subjects' ability to remember the Corsi block sequence.

(Choice A) Sensory memory temporarily stores information from our senses, such as visual or auditory stimuli, just long enough to be transferred to short-term memory. Sensory memory has the shortest duration, from less than 1 second (visual stimuli) to up to 1–2 seconds (auditory stimuli). Sensory memory would not have lasted long enough to aid the subjects in recalling the Corsi block sequence, especially following a distractor task.

(Choice B) Proactive interference occurs when older information interferes with the recall of newer information. If subjects had trouble remembering the Corsi block sequence because of a finger-tapping task learned *the day before*, this would be an example of proactive interference. Episodic memory involves personally relevant information (eg, names of former high school classmates) and was not assessed in this study.

(Choice D) Semantic memory, which was not assessed in this study, involves the recall of facts (eg, names of state capitals).

Educational objective:

Proactive interference occurs when previously learned information interferes with the learning of new information. Retroactive interference occurs when more recent information interferes with the recall of previously learned information.

Time spent:0 sec
Subject:
Behavioral Sciences

QID:400899
System:
Learning, Memory, and Cognition

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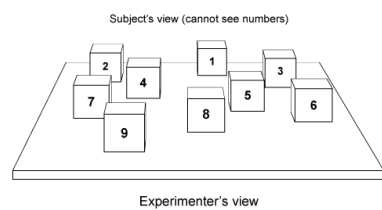


Figure 1 Corsi block-tapping task

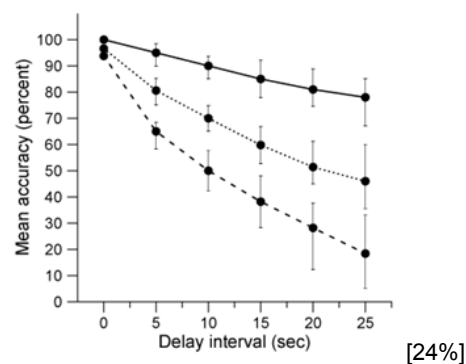
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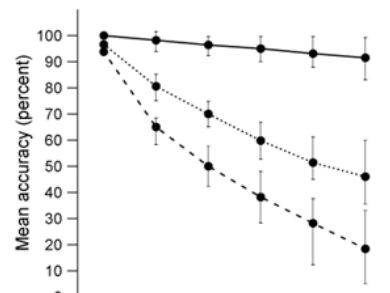
Based on study results and the legend below, which of the following graphs best displays the mean and 95% confidence intervals for accuracy scores for each group?

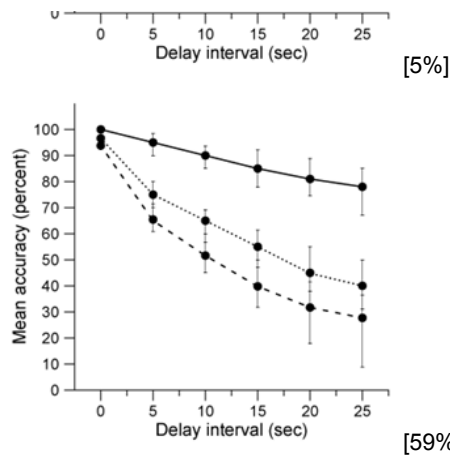
- Control group
-●..... KS group
- - ● - - AD group

☐ A.



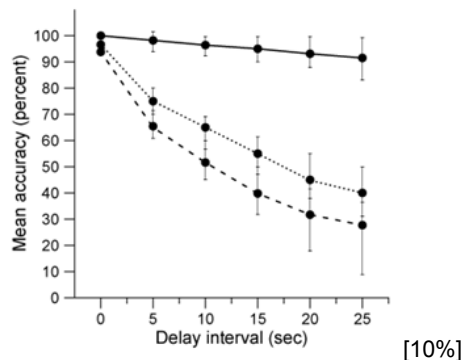
☐ B.





✓ ☒ C.

☐ D.



Correct
59% answered correctly

Explanation:

Error bars may represent several types of statistics (eg, confidence intervals, standard deviations, standard errors). Interpretation of error bars depends on which statistic is depicted.

Confidence intervals (CIs) address how well a sample statistic (eg, mean) estimates a true population value by providing a range of values that is likely to contain the population parameter (eg, mean) of interest. On the graph, the points depict the group mean and the CIs depict the uncertainty of the mean considering the sample size and the variability of the measurement.

Graphically, CIs may help determine statistical significance: Whenever CIs **do not overlap**, group means are statistically **significantly different**. However, the opposite is not always true; whenever CIs **overlap**, group means **may or may not be statistically different**.

This study used a mixed factorial design. It assessed **between-groups** factors by testing **two or more groups simultaneously** to examine differences between groups. It also evaluated **within-groups** (or repeated measures) factors by assessing the **same** subjects at **several time points** to determine if there was a significant change over time.

This study produced two statistically significant findings: (1) the AD and the KS groups were significantly impaired on the delayed memory task *compared to controls* (between-groups finding), and (2) *all subjects* showed a significantly greater mean number of errors as the delay interval increased (within-groups finding).

Therefore, an accurate representation of results should include the following: (1) CIs for the AD and the KS groups that **do not overlap with those for the control group**, and (2) CIs for the final time points (ie, 25-second delay interval) in all groups that **do not overlap with their corresponding initial time points** (ie, 0 seconds).

(Choices A and B) The CIs for the KS and the AD groups do not overlap, suggesting that subjects with AD and KS performed significantly differently *compared to each other*, which was not a finding mentioned in the passage.

(Choice D) Every CI for the control group at each delayed condition included the initial time point (ie, 0 seconds) mean score,

suggesting that performance for controls remained unchanged. This representation contradicts the finding that *all subjects* showed a statistically significant greater mean number of errors as the delay interval increased.

Educational objective:

Between-groups factors compare two or more groups to determine whether there is a difference between group outcome means. Within-groups factors (repeated measures) compare the outcomes for the same group at different times to determine whether there is a change in the outcome. Confidence intervals (CIs) graphically convey the uncertainty of statistics (eg, mean); if CIs do not overlap, the two means are statistically significantly different; however, the opposite is not always true.

Time spent:0 sec
Subject:
Behavioral Sciences

QID:400900
System:
Learning, Memory, and Cognition

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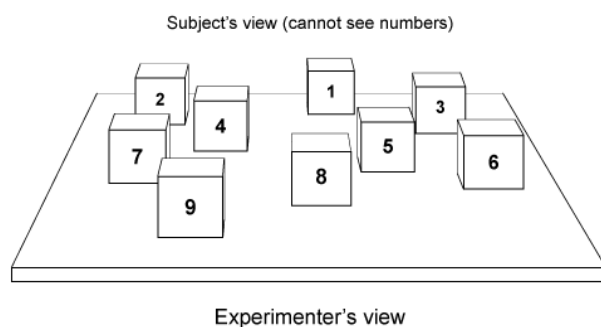


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Matching the healthy control group in terms of age, gender, and premorbid IQ was

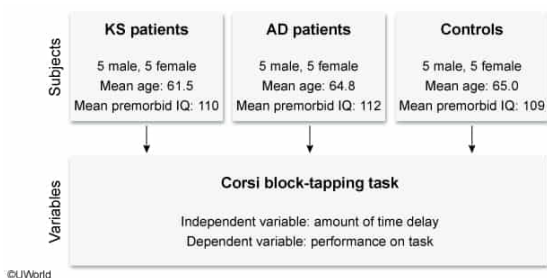
meant to account for all of the following, EXCEPT for the:

- ☐ A. biological factors that affect cognition. [8%]
- ✓ ☒ B. influence of environment on cognitive development. [58%]
- ☐ C. cognitive changes that occur in late adulthood. [5%]
- ☐ D. correlation between types of intelligence. [27%]

Correct

57% answered correctly

Explanation:



Cognitive development is influenced by both heredity and environment. Biological factors affecting cognition include **structural changes** to the brain due to damage or illness and **heredity** (eg, gene and protein expression). **Environmental factors** (eg, drug use) also influence the physiology of the brain, impacting cognition.

The purpose of the study was to assess differences in nonverbal recall in subjects with cognitive dysfunction compared to healthy controls. The control group was matched to the experimental groups in terms of age, gender, and premorbid IQ. **Matching** is meant to **account for confounding variables** that might have an effect on the dependent variable (recall). For example, if the mean age of the control group had been 25, it would have been impossible to determine whether age or time had an effect on recall.

Age, gender, and IQ are all factors that significantly impact cognition, but they can be controlled in an experiment. Subjects' individual experiences with the environment vary widely and are much more difficult, if not impossible, to control. In most human studies, it is difficult to control for environmental factors, such as stress, drug and alcohol use, chemical exposure, social factors, and medical care, because such factors vary widely across individuals.

(Choice A) Matching *does* account for biological factors (eg, age, gender, and IQ) that affect cognition.

(Choice C) Matching *does* account for the cognitive changes that occur in late adulthood (eg, declines in processing speed, memory recall, formation of new episodic memories).

(Choice D) Matching *does* account for premorbid IQ, which often predicts cognitive functioning. Different types of intelligence are correlated, so intelligence must be controlled for in the experiment.

Educational objective:

Matching ensures that only the independent variable has an effect on the dependent variable. Subjects may be matched for biological factors (eg, age, gender, IQ), but it is harder to control for environmental factors, such as stress, chemicals, and social factors.

Time spent:0 sec

Subject:
Behavioral Sciences

QID:400901

System:
Learning, Memory, and Cognition